Comprehensive SOC Incident Response

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1. Executive Summary

This project simulates a real-world cyber attack in a Security Operations Center (SOC) environment. An attacker exploited a vulnerable Samba service using Metasploit. The attack was successfully detected by Wazuh, triaged using TheHive, and mitigated by blocking the malicious IP through CrowdSec. The objective was to demonstrate end-to-end SOC workflow including detection, response, analysis, and reporting.

2. Incident Timeline

16:00 – Attack initiated via Metasploit (Samba exploit)
16:05 – Wazuh generated alert
16:10 – Alert triaged in TheHive
16:20 – Malicious IP blocked via CrowdSec
16:30 – System isolated
17:00 – Post-incident analysis completed

3. Detection and Analysis

Wazuh detected suspicious activity linked to MITRE ATT&CK Technique T1210 (Exploitation of Remote Services). Logs indicated unauthorized access attempts from IP 192.168.1.102. Alert correlation confirmed malicious behavior.

4. Response and Containment

The infected system was isolated from the network. The attacker's IP was blocked using CrowdSec firewall rules. TheHive case management system was used to track and document response actions.

5. Root Cause Analysis (5 Whys)

Why did attack succeed? → Vulnerable Samba service
Why was it vulnerable? → Outdated software
Why not updated? → Lack of patch management
Why no patch policy? → Weak security governance
Why governance weak? → Lack of monitoring and audits

6. Metrics

Mean Time to Detect (MTTD): 2 hours
Mean Time to Respond (MTTR): 4 hours
Dwell Time: 6 hours

7. Recommendations

- Implement regular patch management
- Strengthen monitoring and alerting
- Automate response using SOAR tools
- Conduct user and admin security training
- Perform regular vulnerability assessments

8. Stakeholder Briefing

A cyber attack was detected and successfully mitigated. The attacker exploited a vulnerable service, but quick detection and response minimized damage. Improvements are recommended in patch management and automation to prevent future incidents and enhance security posture.

9. MITRE ATT&CK Mapping

Technique ID	Technique Name	Description
T1210	Exploitation of Remote Services	Used to exploit Samba vulnerability
T1078	Valid Accounts	Unauthorized account access
T1566	Phishing	Initial access simulation